Abdulla Ovais

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.92
C P S INT COLL BHAVAN KHERA HARDOI	Class XII (U.P Board)	2013	72.60%
C P S H S S BHAWANKHERA HARDOI	Class X (U.P Board)	2011	64.00%

PUBLICATIONS

XCOL: xApp-based Framework for Detecting Layer-2 DoS Attacks in 5G NR:

- Contributed to the development of **XCOL**, an **O-RAN-compliant framework** for detecting **Layer-2 DoS attacks**, including **False Base Station (FBS)** and **Malicious User Equipment (MUE)** attacks in 5G NR networks.
- Integrated **SNI5GECT passive air-interface telemetry** with standard **E2 KPM telemetry** from the O-DU and implemented a **Near-RT RIC xApp** for real-time attack detection.
- Developed and evaluated **BiLSTM-Attention and Conv1D sequence models** using **temporal burstiness, RAR suppression ratio, and completion ratio**, achieving **F1 = 1.0** for gNodeB-level MUE flood detection, **F1 = 0.678** for UE-level MUE detection with **100% precision**, and **F1 = 0.922** for FBS-victim identification.

FALCON++: A Semi-Supervised Anomaly Detection and Localization Framework for Resilient O-RAN Deployment:

- Contributed to the development of **FALCON++**, a **semi-supervised anomaly detection and localization framework** designed to improve the resilience of **O-RAN deployments**.
- Implemented a **contextual autoencoder** with **feature-wise gating**, **hierarchical temporal encoding**, and a **causal decoder with residual skip connections** to detect anomalies from temporal telemetry data.
- Used **pairwise CU-DU inferencing** for fault localization and evaluated synthetic faults generated using **STRESS-NG and TC**, achieving **precision = 0.66**, **recall = 0.92**, and **F1-score = 0.77**, outperforming the evaluated baseline methods.

PROJECTS

Secure P2P Payment Platform | PHP, MySQL, Docker, Web Security

- Developed a **secure peer-to-peer payment platform** using **PHP 8.2, MySQL, Apache, and Docker Compose**, implementing user authentication, balance management, and payment transactions.
- Implemented **CSRF protection, secure session cookies, session fixation prevention, password hashing, prepared statements, and login rate limiting** for web application security.
- Built reliable payment processing using **database transactions, row-level locking, balance validation, activity logging, and transaction history**.

Intent-Based Model Selection for Anomaly Detection using ns-O-RAN:

- Built a **telemetry-driven anomaly detection pipeline** using **ns-3/ns-O-RAN** handover traces, analyzing normal and controlled **handover-misconfiguration** scenarios through KPI-based features.
- Trained and evaluated **Random Forest, Gradient Boosting, Logistic Regression, and SVM** models for handover anomaly detection.
- Developed a **rule-based intent-driven model selector** using **accuracy, inference latency, and interpretability** to recommend suitable models based on operator requirements.

Explainable Root Cause Analysis for 5G/O-RAN using RAG & Knowledge Graphs:

- Developed an **LLM-based Root Cause Analysis (RCA)** framework for **5G/O-RAN anomalies** using **Random Forest, SHAP, RAG, and Knowledge Graphs**.
- Built a semantic retrieval pipeline using **Sentence Transformers, Chroma, and Knowledge Graphs** to retrieve relevant **3GPP/O-RAN specifications** for RCA.
- Integrated **DeepSeek/Ollama** to generate explainable root-cause analyses grounded in retrieved telecom-domain knowledge.

5G/O-RAN Telemetry-based Anomaly Detection:

- Developed a **topology-aware anomaly detection pipeline** using **5G/O-RAN CU-DU telemetry**, incorporating temporal and neighboring-node context.
- Evaluated **Random Forest, XGBoost, Isolation Forest, Autoencoders, and GMM** for detecting **CPU stress, memory stress, and network packet-loss anomalies**.
- Performed **per-CU-DU pair and per-fault evaluation** using **Precision, Recall, F1-score, ROC-AUC, and confusion matrices**.

LLM-based MCQ Generation from 3GPP Specifications:

- Developed an **LLM-based MCQ generation pipeline** for creating specification-grounded questions from **3GPP technical documents** using semantic chunking and embedding-based retrieval.
- Implemented **cross-chunk semantic retrieval** using **Sentence Transformers** and local **DeepSeek/Ollama** inference to generate reasoning-oriented questions and distractors.
- Built an automated **self-QA, critique, and refinement pipeline** to improve question correctness and specification grounding.

SKILL

Languages: C/C++, Python, SQL, PHP, JavaScript

Machine Learning & AI: Scikit-learn, PyTorch, XGBoost, Sentence Transformers, RAG, SHAP, LLMs

5G/O-RAN: 5G NR, O-RAN, ns-3, ns-O-RAN, srsRAN, RIC, xApps

Tools & Platforms: Docker, MySQL, Chroma, NetworkX, Ollama, Wireshark, Prometheus, Grafana, OpenSSL, LangChain

SCHOLASTIC ACHIEVEMENTS

- Selected for **Nokia Workshop 2026** on "AI, 6G and Edge-Cloud Convergence," covering emerging technologies in **AI, 6G, and Edge-Cloud systems**.
- Solved **500+ problems** across **GeeksforGeeks, LeetCode, and CSES**.

POSITIONS OF RESPONSIBILITY

Placement Coordinator | Office of Career Services, IIT Hyderabad-Led **recruiter outreach and campus recruitment coordination**, serving as the key liaison between recruiters, students, and the placement office for the 2026-27 hiring season.

RELEVANT COURSES

Relevant Coursework: Machine Learning, Deep Learning, Computer Networks, Wireless Networks

EXTRACURRICULAR

Cricket: Department Champion (**2026**), Department Runner-up (**2025**), Institute Cricket League Runner-up (**2024**), **IIT Hyderabad.**